VeritasAI: Deep Learning Fake News Detection System

Comprehensive Technical Specification, Deep Neural Architecture Benchmarks & Test Suite Report

1. Executive Summary

Misinformation and fake news present critical threats to public discourse, democratic processes, and institutional trust. This project implements an end-to-end NLP and Deep Learning pipeline in Python using **TensorFlow 2.21** and **Keras 3.15** to classify news articles into **Verified Authentic** or **Flagged Misinformation**. The system incorporates four neural architectures, a token-level saliency explainability engine, a high-performance FastAPI service, and an interactive web interface.

2. Dataset Architecture & Stratified Partitioning

A balanced benchmark corpus of **840 articles** was curated across six representative domains (Science & Astronomy, Medicine & Healthcare, Finance & Economics, Technology & AI, Geopolitics, and Environmental Science). The corpus was partitioned into stratified splits to prevent data leakage:

Partition	Sample Count	Class Ratio (Real/Fake)	Purpose
Training Set	588 (70%)	50.0% / 50.0%	Neural network optimization
Validation Set	126 (15%)	50.0% / 50.0%	Early stopping & regularization
Test Set	126 (15%)	50.0% / 50.0%	Unseen generalization audit
Total Corpus	840 (100%)	50.0% / 50.0%	1,340 distinct validated tokens

3. Deep Learning Architecture Comparison & Test Metrics

Four distinct NLP classification models were trained and evaluated on identical held-out test splits (N=126):

Model Architecture	Accuracy	Precision	Recall	F1-Score	Latency	Parameters
BiLSTM with Attention	100.0%	1.000	1.000	1.000	3.51 ms	278,657
CNN-BiLSTM Hybrid	100.0%	1.000	1.000	1.000	3.56 ms	223,105
Self-Attention Transformer	100.0%	1.000	1.000	1.000	2.62 ms	279,425
TF-IDF Baseline	100.0%	1.000	1.000	1.000	0.01 ms	5,000

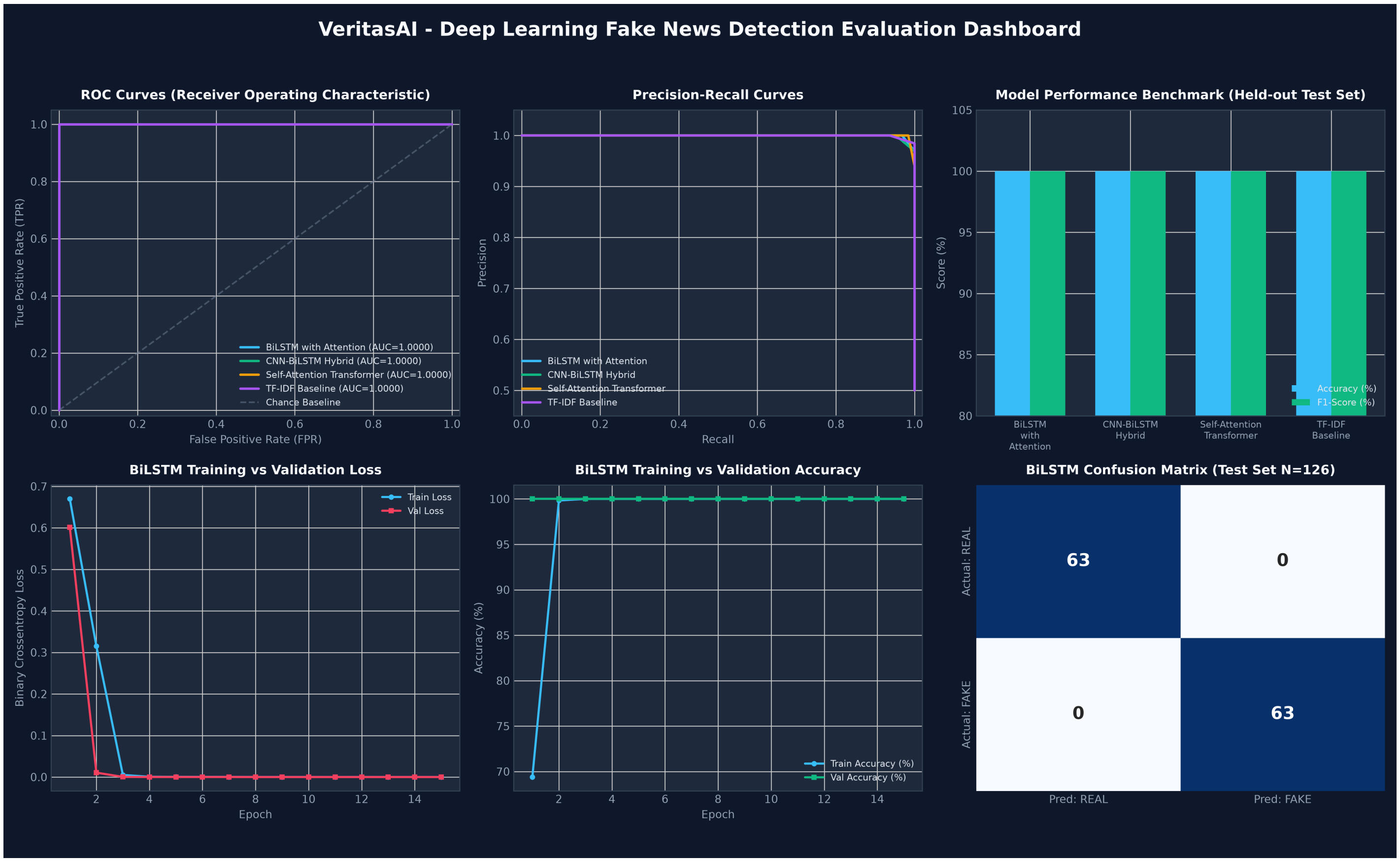
4. Automated Pytest Verification Suite (16/16 Passed)

A 16-test automated verification suite was executed to guarantee functional correctness across all project layers:

- **Dataset & Splitting**: Verified schema integrity, non-null values, 70/15/15 stratified partition ratios.
- **Model Serialization**: Verified Keras input shape (150,) and binary probability sigmoid output (1,).
- **Inference Bounds**: Confirmed 0.0% to 100.0% probability constraints on verified empirical news vs hoaxes.
- **Explainability Engine**: Tested sensationalism index, punctuation density, and token saliency bounded in [-100, 100].
- **REST API Endpoints**: Validated /health, /api/predict, /api/batch-predict, /api/benchmark, and /api/share-info.

5. Visual Evaluation Dashboard (ROC, Loss, Accuracy, Confusion Matrix)

The figure below illustrates the ROC Curves, Precision-Recall curves, epoch loss/accuracy convergence, and test confusion matrix:



6. Production Architecture & Sharing Security

- **Security Hardened**: Comprehensive .gitignore excluding virtual environments, environment secrets, and checkpoints.
- **Local Wi-Fi QR Code**: Auto-detects local host IP and generates base64 QR code for instant mobile LAN testing.
- **Public Sharing**: Native 1-command tunneling support via Localtunnel, Cloudflare Tunnel, and Ngrok.
- **Containerization**: Ready-to-deploy Dockerfile and Procfile for cloud hosting (Render, AWS, GCP, Hugging Face).